

NILM Project - Lab Report: Live-Test Software Improvements

Modeling, Simulation and Automation of Electrical Energy Systems

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Context

Our live NILM system disaggregates per-appliance power from a single PAC4200 measurement, logs every switch event with exact timestamps, and learns unknown devices on the go. It performed well in the live evaluation. This report collects the concrete weaknesses we observed during those tests and the improvements we would prioritise next. Each point states what the software does today, why the current behaviour is a limitation, and the proposed change.

1. A graphical event timeline

Today. Switch events are written to events.csv and shown on the dashboard as a scrolling text log. The measured power is charted, but the on/off history of individual devices is not.

Limitation. During and after a test it is hard to reconstruct what ran when. Reading a text log line by line does not show overlap between devices, missed edges, or a claim that flickered, all of which are obvious at a glance on a timeline.

Improvement. Render a per-device Gantt-style timeline (one lane per device, coloured on intervals against a shared time axis) alongside the existing power chart. The event log already carries the exact edge timestamps and the matched device, so no new data is needed, only the view. It would also serve as a post-test artifact for the report and for debugging misclassifications.

2. Faster and more precise corrections

Today. When a device's steady state changes shape, the engine can briefly misread it before a second evidence source corrects it. In the live test at about 16:01 the PV export was released as off for roughly ten seconds, then the residual monitor re-matched its signature and restored it to on.

Limitation. A PV generation ramp is a step of delta P with delta Q near zero, which is indistinguishable in the (P, Q) plane from a small switching supply turning off, so the off-edge wrongly released the PV claim. The correction is real but slow, because it waits for the residual monitor's multi-second window to accumulate before re-matching.

Improvement. Move the discriminating evidence to the moment of the edge instead of waiting for the residual monitor. A true switch carries the device's harmonic current with it; a generation ramp carries none. Applying that harmonic and sign test at edge time, plus a short hold before a generator claim is released, would shorten the ten-second correction to a fraction of a second and avoid the visible flicker, rather than detecting and undoing the mistake after the fact.

3. Generic load-class labels for unknown devices

Today. Teaching an unknown load requires the user to type a specific name (standing_fan_high, coffee_machine_standby, and so on). The signature table and the identify model are then keyed to that exact family.

Limitation. In a real installation the operator often does not know the device family, only that something new is drawing power. Requiring a specific name makes the system brittle to naming and unhelpful when the device is simply unfamiliar.

Improvement. Offer an automatic electrical-class label derived from features we already measure, so an unknown load can be reported as resistive, inductive, capacitive, or non-linear (switch-mode) without a device name. The sign of Q separates inductive from capacitive; power factor and THD_I separate a clean resistive heater from a distorted electronic supply. Our own fingerprint table shows these classes are cleanly separable on this hardware (THD_I of about 2 percent for the boiler versus about 170 percent for the laptop). The user could still refine the auto-label into a specific name later, but the system would be useful immediately.

4. Fully asynchronous, hands-off training

Today. The in-mix teach removed the need to empty the mains, but it still asks the user for one manual action: switch the unknown device off and back on once so the settled step gives a model-free measurement of its draw.

Limitation. Any required user action interrupts an experiment and is impossible for loads that cannot be toggled (a running fridge, an inverter, a sealed installation). It also prevents unattended, continuous learning.

Improvement. Learn passively from the residual history the engine already records, with no toggle required. When unexplained power appears and stays, capture its settled level, Q , and harmonic signature over time and teach a provisional signature from that alone, cross-checking against any natural on/off transitions that happen to occur. The manual toggle would become an optional way to improve confidence, not a prerequisite, making training a background process the user never has to trigger.

5. Recognition across a device's operating range

Today. A device is taught at whatever it happened to draw during the capture. A switching supply or PV inverter whose output varies with load or sunlight can then fall outside the taught range and stop matching (the laptop taught at 66 W idles at 43 W; PV taught around -13 W was observed at -30 W).

Limitation. Variable-output loads and generation are exactly the cases NILM finds hardest, and a single taught operating point does not cover them.

Improvement. Teach a range rather than a point: record the device across several operating levels, or let a distinctive harmonic fingerprint taper the match toward idle (already done for high-THD loads). For PV specifically, the unique and load-independent discriminator is the sign of power, which no consumer produces, so generation should be recognised by sign across its whole range rather than by a taught wattage.

6. Summary

The system met the goals of the exercise. The improvements above target the gaps we felt during live operation: better visibility (timeline), faster self-correction (edge-time harmonic evidence), lower operator burden (generic labels and hands-off learning), and robustness to the hardest loads (variable-output devices and generation). None require new sensing hardware; all build on features the pipeline already records.